

Introductory Biology

Science

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Introductory Biology (A07888)

Short Title: Introductory Biology
Faculty Science and Computing
Department Science
Cluster Biology
Author ROMA HONY
Credits 5

Level: Introductory

Description of Module / Aims

The aim of this module is to introduce the student to the fundamentals of biology. Basic cell structure, diversity of organisms, introductory genetics and ecology will be covered, together with relevant supporting practical skills. This module is designed to be delivered across all laboratory based 1st year science programmes.

Programmes

BIOL-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
BIOL-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
BIOL-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
BIOL-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
BIOL-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
BIOL-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
BIOL-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
BIOL-0001	BSc in Agricultural Science (WD_SAGRI_D)	1 / 1 / M
BIOL-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
BIOL-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
BIOL-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Prokaryotic and Eukaryotic basic cell structure
- Nutrition and culture of microorganisms
- History, biology and diversity of organisms
- Microscopy and staining techniques
- Cell reproduction and introduction to genetics
- Introduction to ecology

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain basic microbial cell structure and reproduction.
2. Recognise the diversity of organisms and distinguish between major groups of organisms.
3. Use standard laboratory techniques to demonstrate aseptic technique, basic microscopy and staining skills and other relevant practical work.
4. Describe the fundamentals of ecology.
5. Outline key concepts in genetics at an introductory level.
6. Record results of laboratory work using worksheets.
7. Record results of tutorial work using an online learning platform (Moodle).

Learning and Teaching Methods

- Introductory biology is delivered in a multi-modal approach.
- The module consists of lectures, laboratory practicals and online tutorials (on- or off-site).
- The teaching strategy is to adopt a blended learning approach. Lecture sessions are delivered using a flipped classroom style.
- Resource material is made available to the student prior to the lecture session. The session is delivered in a non-didactic manner.
- The student completes e-activities in the form of quizzes which are designed pre- and post-lecture sessions.
- Audio and video files of module content are provided to cater for the variety of learner types (class size < 50).
- Laboratory practicals and tutorial sessions are linked.
- Pre-practical quizzes are used as an formative learning method and these can be completed during tutorial session on or off-site.
- All of the above are delivered through Moodle e-learning platform.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Practical</i>	50%	3,6
<i>Tutorial/Problem Sheets</i>	20%	1,7
<i>In-Class Assessment</i>	30%	2,4,5

Assessment Criteria

<40%: No understanding of basic biological concepts. Unable to carry out practical skills in laboratory.

40%-49%: Some understanding of biological concepts and has basic level of competency in the required skills in microbiology.

50%-59%: A clear understanding of the biological concepts presented. Good practical skills.

60%-69%: In addition to the above, shows good critical thinking skills, engages in debate in class and has a high level of efficiency in the microbiology lab.

70%-100%: All of the above to an excellent level.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	8	
Practical	16	
Independent Learning	73	

Essential Material

- "Introductory Biology module Moodle page." <http://www.moodle.wit.ie>

Supplementary Material

- Madigan, M., J. Martinko, K. Bender, D. Buckley and D. Stahl. *Brock Biology of microorganisms*. 14th ed. Boston: Pearson Education; Benjamin-Cummins, 2014.

Requested Resources

- Room Type: Biology Lab
- Lecture Room: Loose Seated
- Room Type: Computer Lab